

Stochastic Process

From Kurt Jacobs' Stochastic Processes for Physicists.

Chapter 3: Stochastic Equations with Gaussian Noise

3.1 The Continuum Limit and Wiener Noise

To describe a system driven by random Gaussian increments, we start with a discrete difference equation: $\Delta x = f(t)\Delta t + y_n\Delta t$. Taking the limit $\Delta t \rightarrow 0$ leads to the Wiener process $W(t)$, where the infinitesimal increment dW is a Gaussian random variable.

- Variance Scaling: For a well-defined continuum limit where the variance of $W(T)$ is proportional to T , we are forced to set the variance of the increment dW equal to dt ($VdW = dt$).
- Order of Noise: Because $VdW = dt$, the standard deviation is $\sqrt{dt}$. This means dW is of order $\sqrt{dt}$, which is larger than dt in the infinitesimal limit.

3.2 Ito's Rule: $(dW)^2 = dt$

A fundamental departure from standard calculus is that the square of the noise increment does not vanish.

- Proof by Variance: The expectation value $\langle (dW)^2 \rangle = dt$. The variance $V \sum (dW_n)^2$ scales as $1/N$ and vanishes as $N \rightarrow \infty$
- Result: In the continuum limit, the sum of $(dW)^2$ is deterministic: $(dW)^2 = dt$. All other higher-order products like $dt dW$ or $(dt)^2$ vanish

3.3 Ito's Formula (Change of Variables)

To change variables from x to $y(x, t)$, a second-order Taylor expansion is required because $(dx)^2$ contains a term proportional to $(dW)^2 = dt$.

$$dy = \frac{\partial y}{\partial x} dx + \frac{\partial y}{\partial t} dt + \frac{1}{2} \frac{\partial^2 y}{\partial x^2} (dx)^2$$

For $dx = fdt + g dW$, substituting dx and discarding terms higher than dt yields:

$$dy = \left(f \frac{\partial y}{\partial x} + \frac{\partial y}{\partial t} + \frac{g^2}{2} \frac{\partial^2 y}{\partial x^2} \right) dt + g \frac{\partial y}{\partial x} dW$$

3.4 Key Analytical Solutions

- Ornstein–Uhlenbeck (OU) Process: $dx = -\gamma x dt + g dW$. Solution: $x(t) = x_0 e^{-\gamma t} + g \int_0^t e^{\gamma(s-t)} dW(s)$ 12, 15. This is a Gaussian process with mean $x_0 e^{-\gamma t}$ and steady-state variance $g^2/2\gamma$
- Full Linear SDE: $dx = -\gamma x dt + g x dW$. Solution: $x(t) = x_0 \exp -(\gamma + g^2/2)t + gW(t)$

Chapter 4: Further Properties of Stochastic Processes

4.1 Sample Paths and Differentiability

- Non-differentiability: Because $dW/dt \sim 1/\sqrt{dt}$, the derivative of a Wiener process tends to infinity as $dt \rightarrow 0$. Thus, sample paths are nowhere differentiable.
- Non-anticipating functions: Functions $f(t)$ are "adapted" if they only depend on noise increments dW up until time t .

4.2 First-Passage Time and Reflection Principle

The time T_a to hit boundary a is found by noting that for every path crossing a and ending above a , there is an equally likely reflected path ending below a

- Probability Density: $P(T) = \frac{a}{\sqrt{2\pi}} \frac{e^{-a^2/(2T)}}{T^{3/2}}$
- Mean: The average time to reach a boundary in an unbounded space is infinite ($\langle T_a \rangle = \infty$)

4.3 Correlation and White Noise

- Auto-correlation: For a Wiener process, $\langle W(t)W(t+\tau) \rangle = \min(t, t+\tau) = t$
- White Noise ($\xi(t)$): Defined as the formal derivative dW/dt . It is an idealization with a delta-function correlation $\langle \xi(t)\xi(t+\tau) \rangle = \delta(\tau)$ and a flat power spectrum $S(\nu) = 1$

Chapter 5: Applications of Gaussian Noise

5.1 Brownian Motion (Langevin Model)

A particle of mass m in a fluid: $dp = -\gamma p dt + g dW$.

- Thermalization: To match statistical mechanics ($\langle p^2/2m \rangle = k_B T/2$), the noise strength must be $g = \sqrt{2\gamma m k_B T}$ 16, 30.
- Long-time limit: For $t \gg 1/\gamma$, the variance of position $Vx(t)$ is proportional to t , matching the Wiener process behavior.

5.2 Finance: Black–Scholes Equation

Stock prices are modeled as $dS = \mu S dt + \sigma S dW$.

- Delta Hedging: By creating a portfolio $\Pi = C + \alpha S$ and choosing $\alpha = -\partial C / \partial S$, the noise dW cancels out, making the portfolio risk-free.
- PDE: Equating the risk-free return to the interest rate $r\Pi dt$ yields the Black–Scholes PDE:

$$\frac{\partial C}{\partial t} + rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} - rC = 0$$

5.3 Stratonovich vs. Ito Integrals

- Stratonovich ($\circ dW$): Evaluates the integrand at the midpoint of the interval 37, 38. It obeys standard calculus rules but is harder to solve.
- Conversion: Physical noise is often Stratonovich. To use Ito tools, convert using the drift correction: $dx = (f + \frac{1}{2}g \frac{\partial g}{\partial x})dt + g dW$.

Chapter 6: Numerical Methods

- Euler Method: $\Delta x = f\Delta t + g\Delta W$. It has strong order 0.5 (error $\sim \Delta t$) and weak order 1.0 (for moments).
- Milstein Method: Improves accuracy to strong order 1.0 by adding the Ito correction term:

$$\Delta x = f\Delta t + g\Delta W + \frac{1}{2}g \frac{\partial g}{\partial x} (\Delta W)^2 - \Delta t$$

- Stability: For "stiff" equations with fast oscillations, implicit methods (evaluating f at $t + \Delta t$) are used to prevent exponential error growth.

Chapter 7: Fokker–Planck Equations (FPE)

7.1 Derivation and Current

The FPE describes the evolution of the probability density $P(x, t)$ 48. For an SDE $dx = fdt + \sqrt{D}dW$:

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x} f(x, t)P + \frac{1}{2} \frac{\partial^2}{\partial x^2} D(x, t)P$$

49. This is a continuity equation $\frac{\partial P}{\partial t} + \frac{\partial J}{\partial x} = 0$, where $J = fP - \frac{1}{2} \frac{\partial(DP)}{\partial x}$ is the probability current.

7.2 Boundaries and Stationary Solutions

- Absorbing: $P(c, t) = 0$ 51.
- Reflecting: $J(c, t) = 0$ 52.
- 1D Steady State: If $J = 0$ everywhere, $P(x) = \frac{1}{ND(x)} \exp\left(\int \frac{2f(u)}{D(u)} du\right)$

7.3 Exit Times via the Backward FPE

The probability of remaining in an interval a, b obeys the backward FPE (acting on initial coordinates y):

$$\frac{\partial P_{in}}{\partial t} = f(y) \frac{\partial P_{in}}{\partial y} + \frac{1}{2} D(y) \frac{\partial^2 P_{in}}{\partial y^2}$$

Chapter 8: Jump Processes

8.1 Poisson Process and Master Equation

Driven by discrete increments $dN \in \{0, 1\}$ with $P(dN = 1) = \lambda dt$.

- Poisson Distribution: $P(n, t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$
- Jump Rule: $(dN)^2 = dN$
- Master Equation: The discrete version of FPE: $\frac{dP(n, t)}{dt} = \lambda P(n-1, t) - \lambda P(n, t)$

8.2 Generating Functions

Defined as $G(s, t) = \sum s^n P(n, t)$. This converts the master equation into a PDE for G , from which moments are easily derived: $\langle N \rangle = \frac{\partial G}{\partial s} \big|_{s=1}$.

8.3 Applications

- Predator-Prey: Modeled by jump rates for births, deaths, and hunting.
- Stochastic Resonance: In neurons, adding noise can actually improve the detection of a weak signal by regularizing jump intervals.

Chapter 9: Levy Processes

9.1 Stable Distributions

Levy processes are the most general i.i.d. increments.

- α -stable index: Processes are characterized by $\alpha \in (0, 2]$. Only $\alpha = 2$ (Gaussian) has a finite variance 68, 69.
- Scaling: The width of the density scales as $t^{1/\alpha}$. For the Cauchy process ($\alpha = 1$), width scales linearly with t .

9.2 Levy–Khinchin Representation

Every Levy process is a combination of:

1. Deterministic Drift (dt)
2. Gaussian Diffusion (dW)
3. Jump Processes (dN)

4. Infinite Activity Jumps (infinite rate of infinitesimal jumps)

9.3 Levy Calculus

The transformation formula for $y(x, t)$ includes a jump summation term:

$$dy = (\dots)dt + (\dots)dW + \sum y(x + \Delta J) - y(x)$$